



CPSC 100

Computational Thinking

Data Visualization

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia



Agenda

- Course Admin
 - Reminder about Milestone 5
- Mini-activity on the importance of Data Visualizations
- Gestalt Principles

Course Admin

Grading Scheme

Item	Weight	Due date(s)	Category
Learning Logs (10)	7%	Fridays at 6 PM	Homework
Labs (9)	18%	Fridays at 6 PM	Homework
Term Project (5)	25%	TBD	Project
Tests (10)	50%	Weekly	Exam

! Attention

Due dates for Learning Logs, Labs, and the Term project have an automatic 48 hour grace period after the due dates listed above. Any submissions submitted past the grace period will not be graded (unless prior arrangements have been with the instructor before the due date.).

Passing requirements

- Students must satisfy ALL conditions to pass the course:
 1. Pass the Lab component with a grade of at least 50%,
 2. Pass your seven (7) bests Tests (together) with a score of at least 50%,

If a student does not satisfy these requirements, the student may be assigned the **lower** of their earned course grade or 45% in the course.

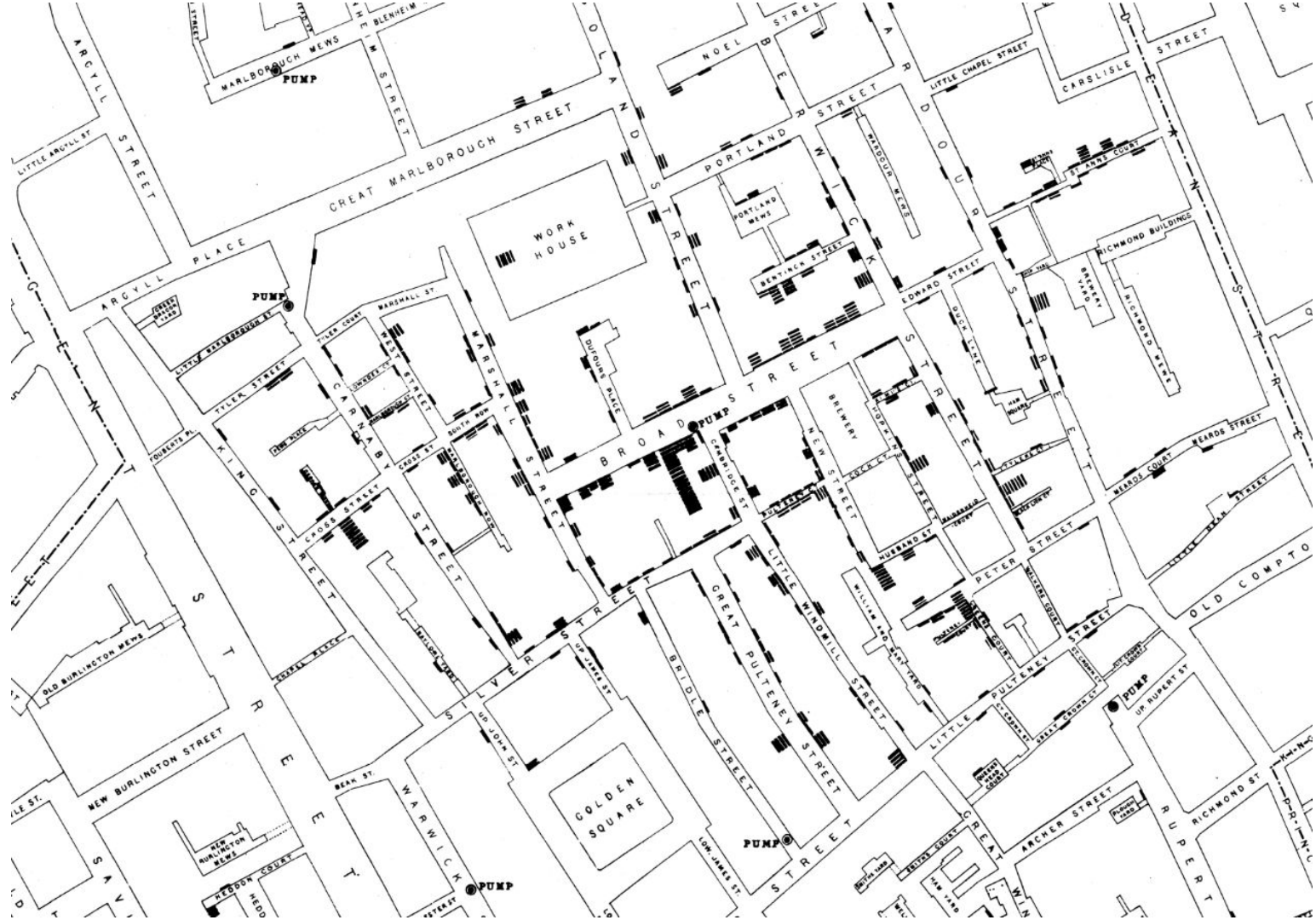
Mini-Activity

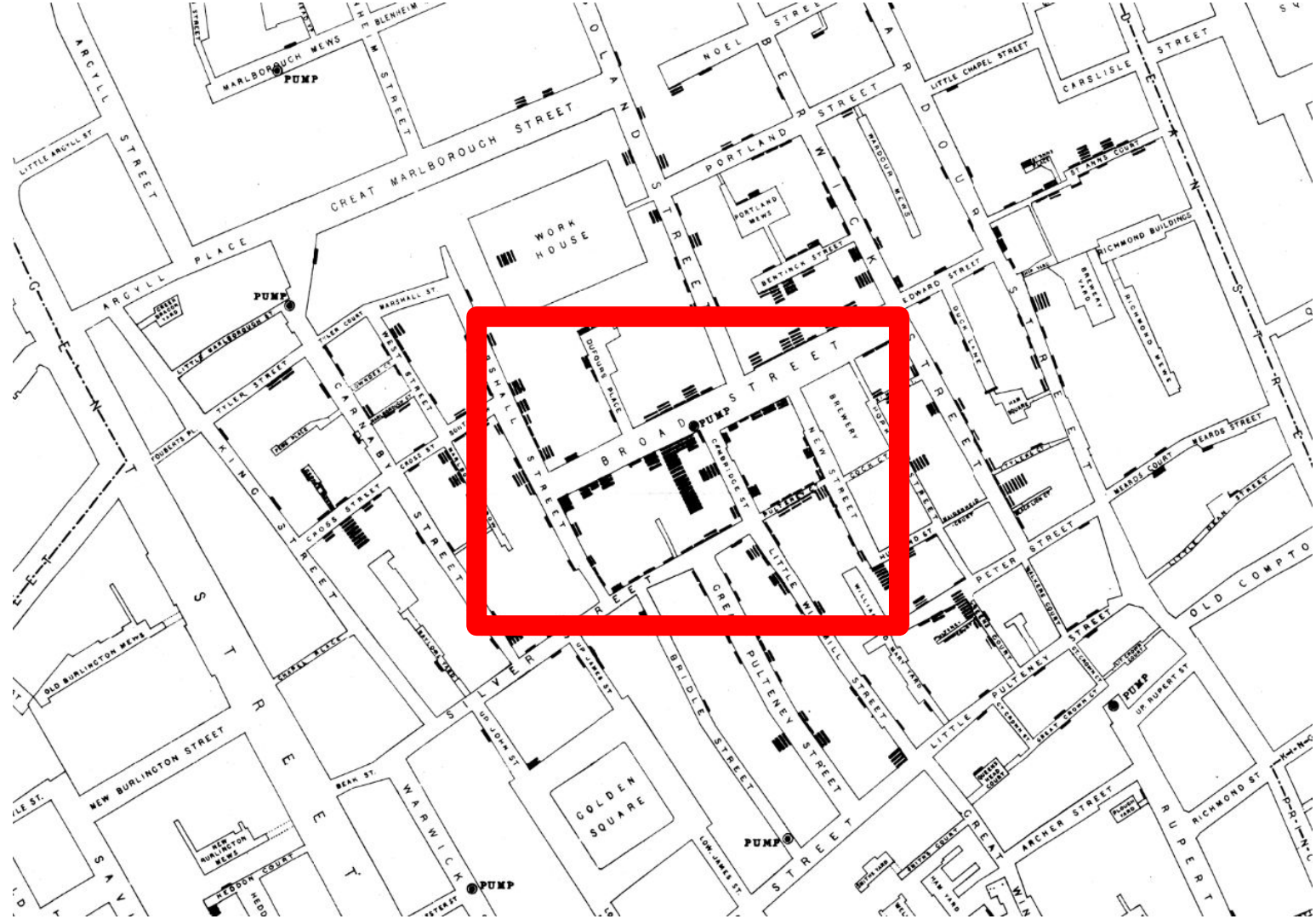
In groups of 3-4: Solve the Cholera Outbreak!

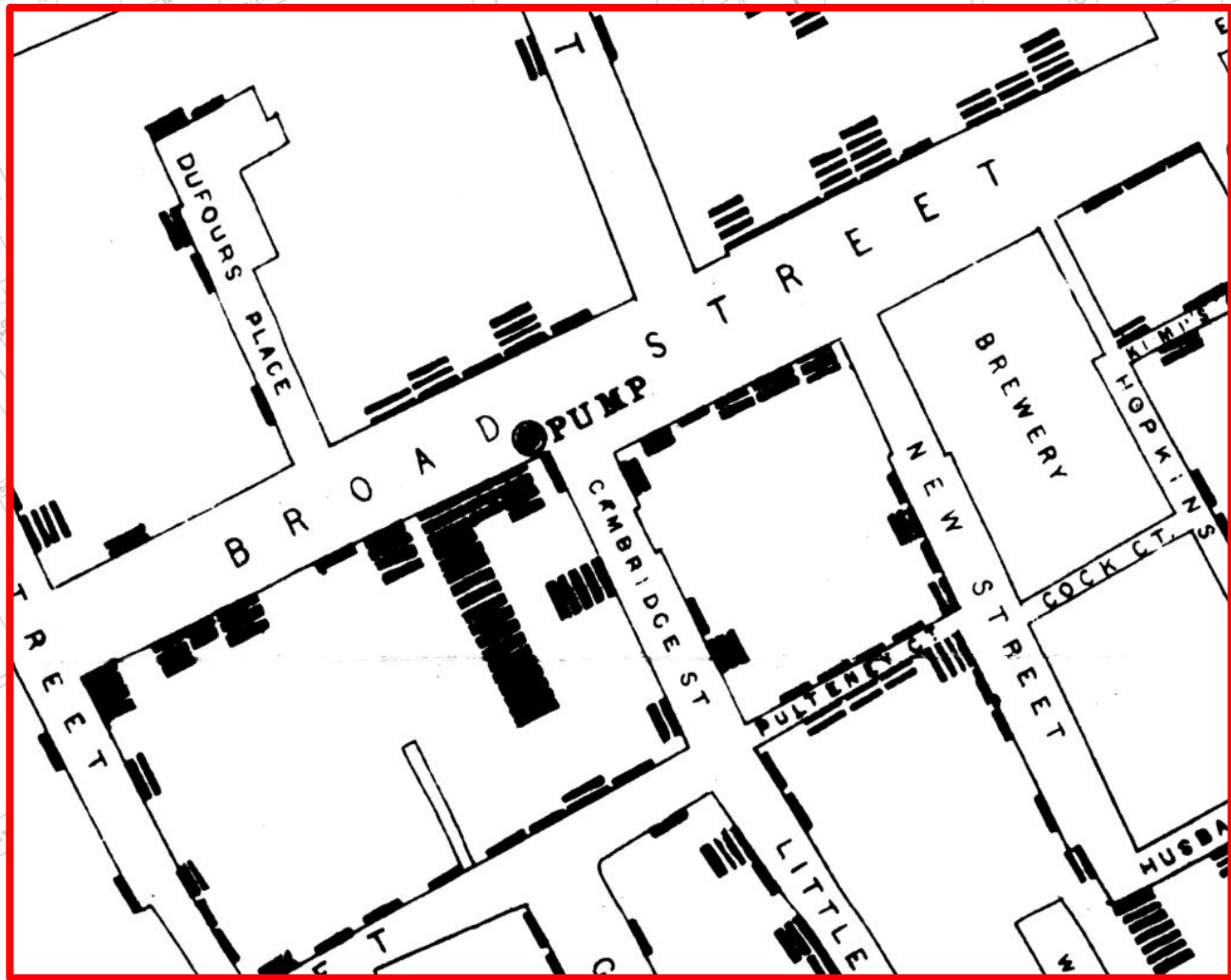
- It is **1854** (remember: no computers!) and the government has told you to find the source of the outbreak.
 - What would you do?
 - What kind of information would you want to collect?
 - What would you do with the information? Since there are no computers help organize information, how would you deal with the amount of information that you get?



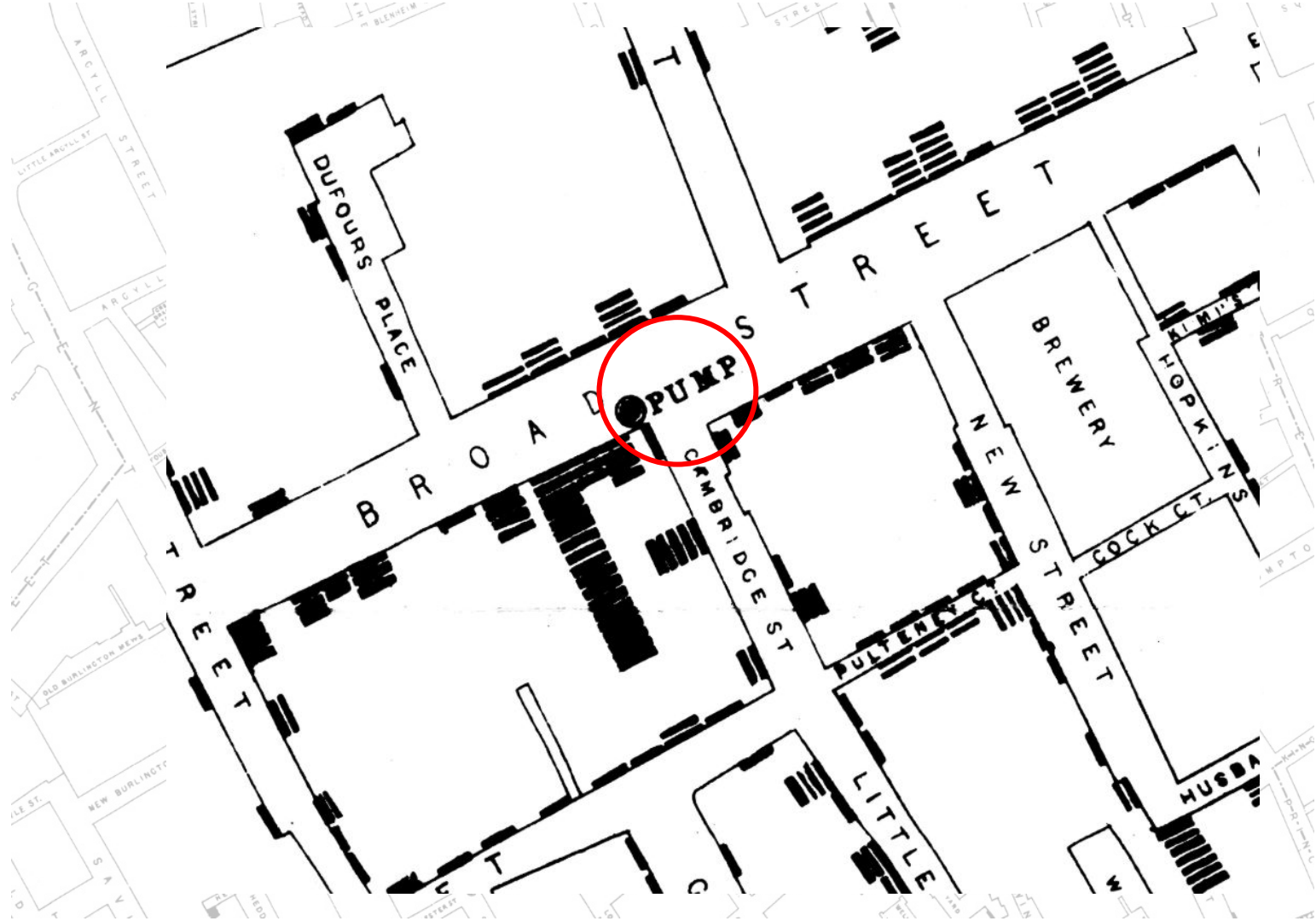
A COURT FOR KING CHOLERA.











Learning Goals



Learning Goals

After this **today's lecture**, you should be able to:

- Understand, define, name Gestalt Principles and explain their role in visualization
 - Demonstrate how these principles appear in real life—such as recognizing how proximity is used in store displays or how logos use closure to create memorable designs.
- Explore how understanding these principles can inform better visual communication, whether for creating infographics, arranging elements on a webpage, or organizing a physical space.

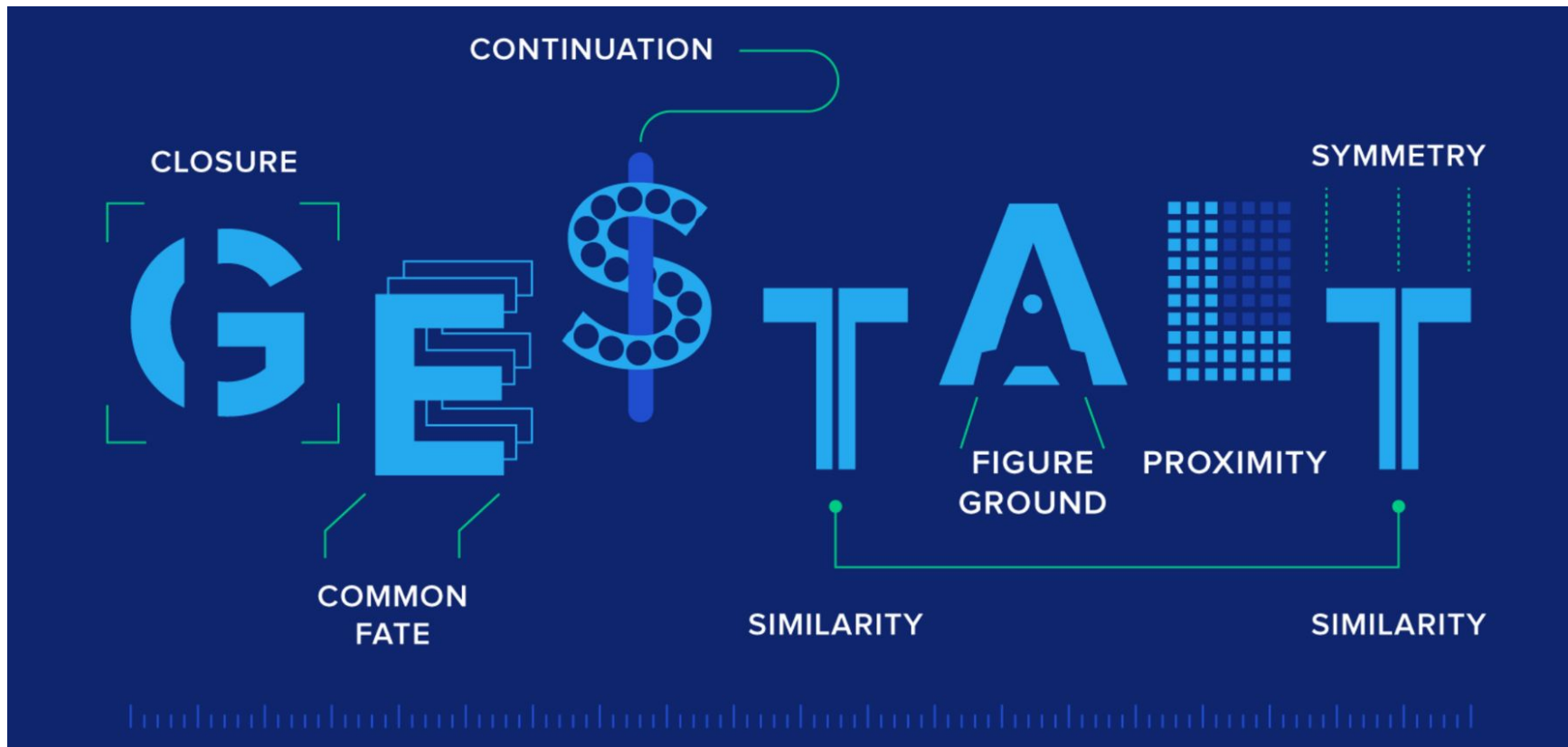
Gestalt Principles



What are Gestalt Principles?

- "Gestalt" is German for "unified whole".
- German psychologists Max Wertheimer, Kurt Koffka, and Wolfgang Kohler created the **Gestalt Principles** in the 1920s.
- They wanted to understand **how people make sense of the confusing things they see and hear**. They identified a set of laws that address the natural compulsion to find order in disorder. According to this, the mind "informs" what the eye sees by perceiving a series of individual elements as a whole.

What are Gestalt Principles?





Gestalt Principles

1. Emergence
2. Closure (Reification)
3. Common Region
4. Continuity or Continuation
5. Proximity
6. Multistability
7. Figure/Ground
8. Invariance
9. Pragnanz
10. Similarity
11. Symmetry and Order
12. Common Fate



Today's Focus

1. Emergence
2. Closure (Reification)
3. Common Region
4. Continuity or Continuation
5. Proximity
6. Multistability
7. Figure/Ground
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Emergence

Gestalt Rule: **Emergence**



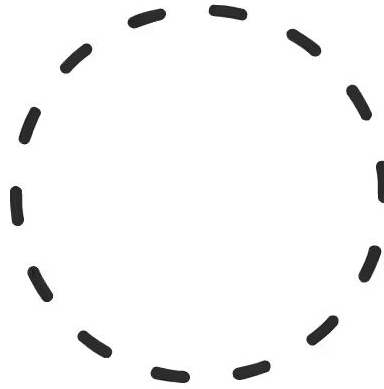
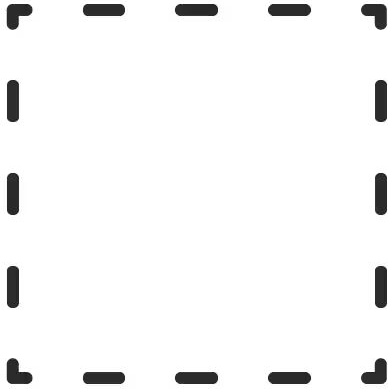
Emergence

- A dog can immediately be identified, despite the picture not having all the details.
- Instead of building up our perception piece by piece, our minds tend to see the overall form all at once.
- We perceive the world without thinking too much about understanding every small thing around us.



Closure

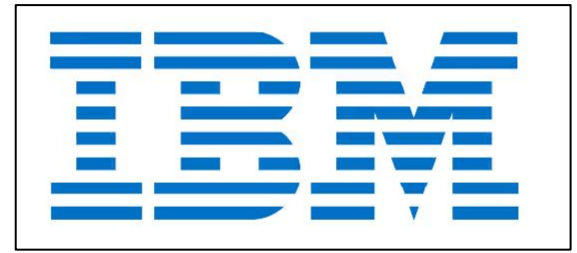
Gestalt Rule: **Closure**





Closure

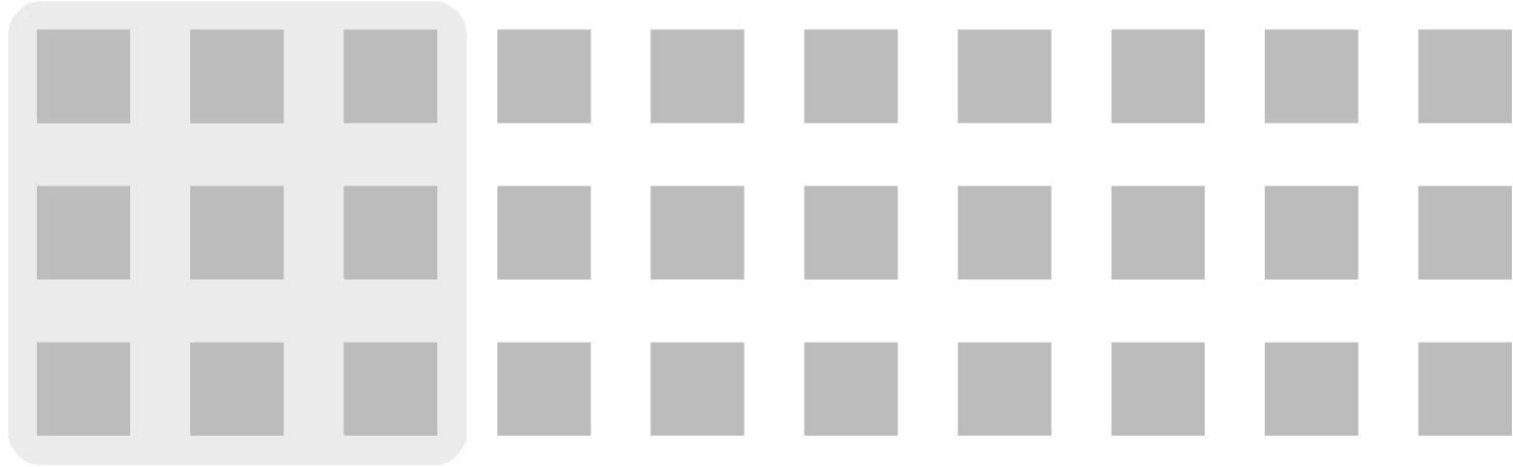
Sometimes known as "Reification"



- The mind's tendency to fill in gaps in incomplete shapes or figures, effectively "closing" open boundaries to perceive a whole.
- Lets us quickly recognize objects even when parts are missing.
- Used in logos and artwork to hint at forms and engage the viewer's imagination.

Common Region

Gestalt Rule: **Common Region**



Common Region

- Elements located within the same clearly defined area or boundary are perceived to belong together.
- Helps us group items based on shared space.
- Influences how we arrange or label items so they appear unified (like sections on an infographic or webpage).
- In a living room, a rug can create a sense of a "zone" or common elements are on facebook posted





Today's Focus

1. Emergence

2. Closure (Reification)

3. Common Region

4. Continuity or
Continuation

5. Proximity

6. Multistability

7. Figure/Ground

8. Invariance

9. Pragnanz

10. Similarity

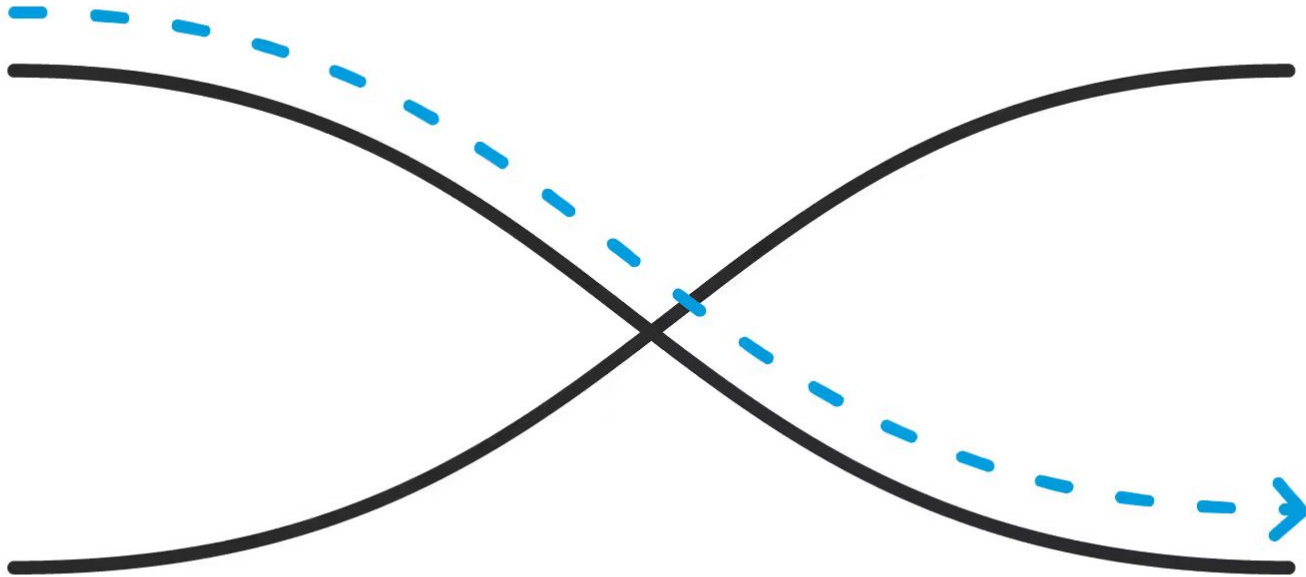
11. Symmetry and Order

12. Common Fate



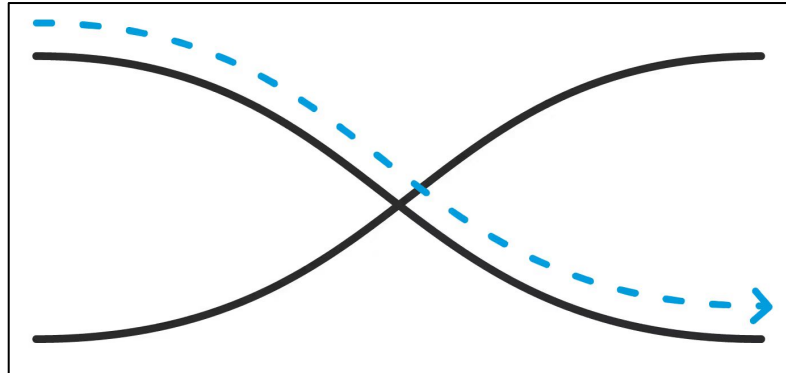
Continuity

Gestalt Rule: **Continuity**



Continuity

- Our eyes prefer continuous, unbroken lines and curves. When lines or shapes follow a smooth path, we assume they are connected, rather than separate elements.
- Makes complicated or intersecting shapes **easier** to interpret.
- Encourages design that “**flows**,” guiding your attention **naturally**.



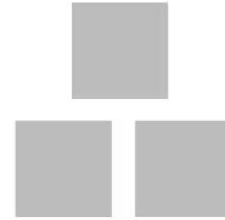
Continuity

- <https://youtu.be/gVIUI1832CE?si=IMhrru9UyeWmVJQi>



Proximity

Gestalt Rule: **Proximity**





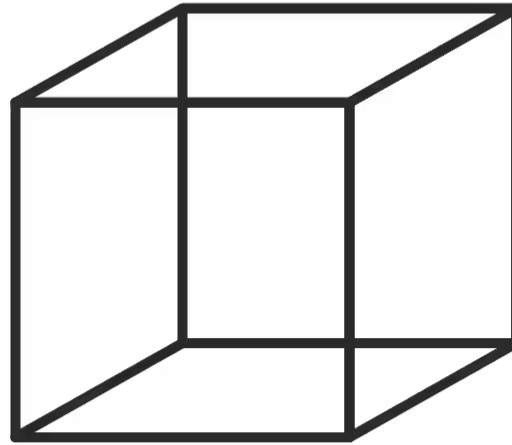
Proximity

- Objects that are physically close to one another are seen as related or forming a group.
- Helps with quick organization of information.
- Influences how we arrange items to signal relationships (e.g., grouped shelves in a grocery store).



Multistability

Gestalt Rule: **Multistability**



Necker Cube



Multistability

- Phenomenon where an ambiguous image can switch between different interpretations in your mind.
- Demonstrates how our brains can hold multiple interpretations of the same visual.
- Highlights the importance of context and focus in shaping what we perceive first.

"My wife and my mother-in-law" from 1915 by W. E. Hill



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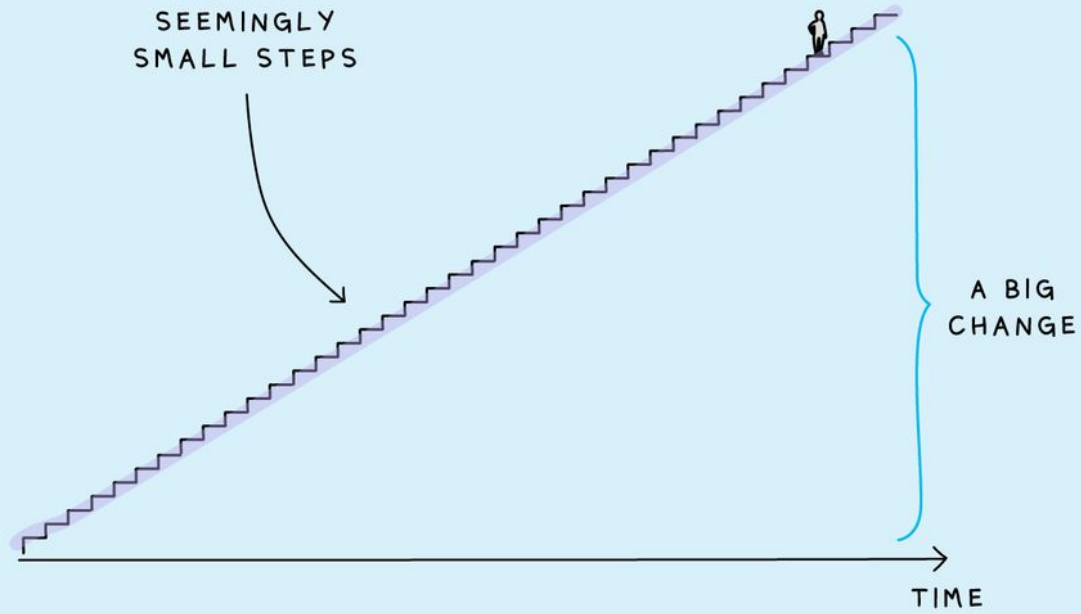




Gestalt Principles (Part 1)

- 1. Emergence**
- 2. Closure (Reification)**
- 3. Common Region**
- 4. Continuity or Continuation**
- 5. Proximity**
- 6. Multistability**
7. Figure/Ground
8. Invariance
9. Pragnanz
10. Similarity
11. Symmetry and Order
12. Common Fate





LIZ FOSSLIE



Q: Which Gestalt Principles is most applicable to this logo?



- A. Emergence
- B. Closure
- C. Common Region
- D. Continuity
- E. Multistability



Course Retrospective

Learning Goals

At the highest level, this courses has three main goals.

Students who complete this course will be able to

- explain foundational computing concepts,
- explore applications built on these foundations,
- describe the implications of the applications.



What this course includes



Algorithms &
Programming

Data &
Internet

AI &
Data Mining

Data
Visualization

Week	Week of	Topic
1	Jan 5	Introductions
2	Jan 12	Algorithms I
3	Jan 19	Algorithms II
4	Jan 26	Data Representation
5	Feb 2	Programming I
6	Feb 9	Programming II
7	Feb 16	Reading week
8	Feb 23	Internet and HCI
9	Mar 2	Artificial Intelligence I
10	Mar 9	Artificial Intelligence II
11	Mar 16	Catch-up week
12	Mar 23	Data Mining
13	Mar 30	Data Visualization
14	Apr 6	Final Projects



DECOMPOSITION

Breaking down problems into smaller, easier parts.



PATTERN RECOGNITION

Using patterns in information to solve problems.



ABSTRACTION

Finding information that is useful and taking away any information that is unhelpful.



MODELLING AND SIMULATION

Trying out different solutions or tracing the path of information to solve problems.



ALGORITHMS

Creating a set of instructions for solving a problem or completing a task.



EVALUATION

Assessing a solution to a problem and using that information again on new problems.



Q: Which Gestalt Principles is most applicable to this logo?



A. Emergence

B. Closure

C. Common Region

D. Continuity

E. Multistability





Q: Which Gestalt Principles is most applicable to this logo?



- A. Emergence
- B. Closure
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Q: Which Gestalt Principles is most applicable to this logo?



- A. Emergence
- B. Closure
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- E. Multistability

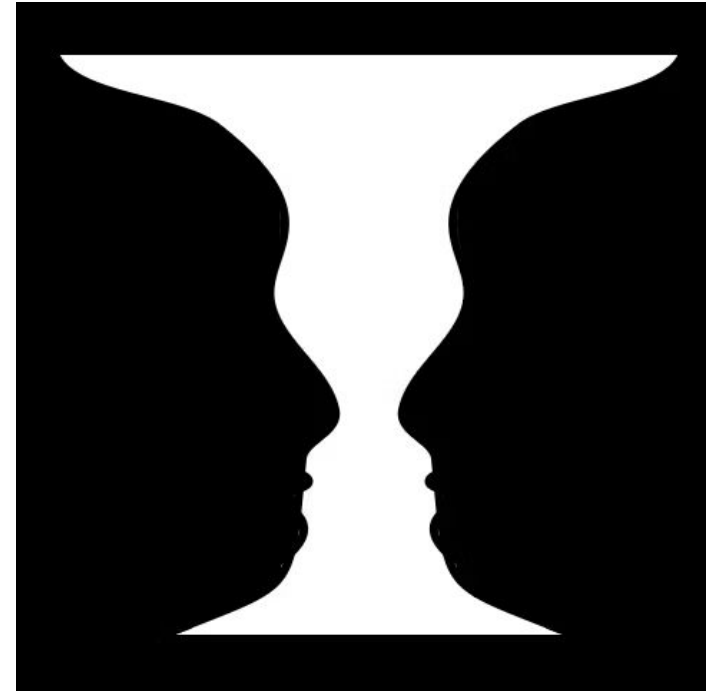




Q: Which Gestalt Principles is most applicable to this image?



iClicker



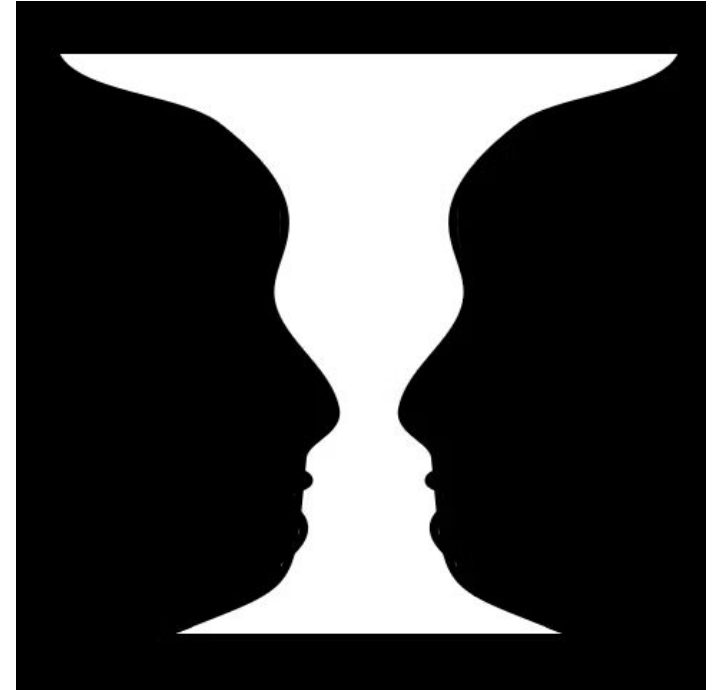
- A. Emergence
- B. Closure
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Q: Which Gestalt Principles is most applicable to this image?



iClicker

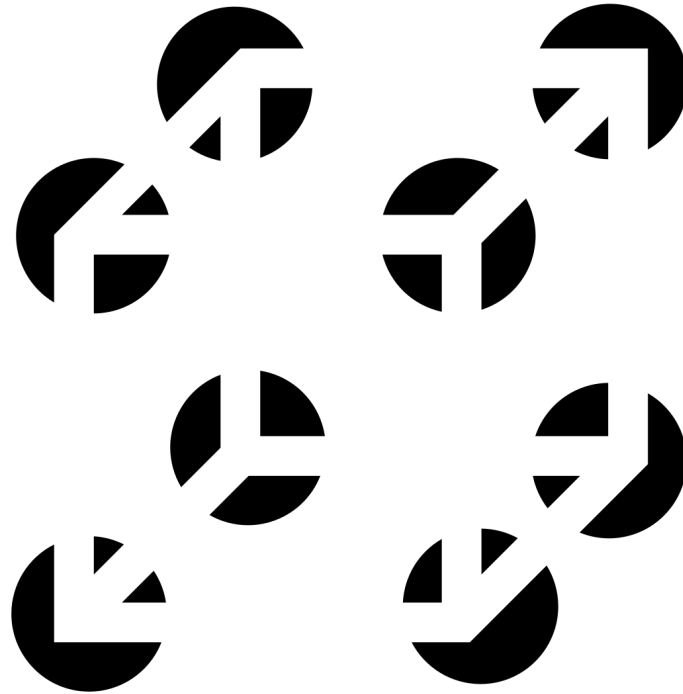


- A. Emergence
- B. Closure
- C. Common Region
- D. Continuity
- E. Multistability**

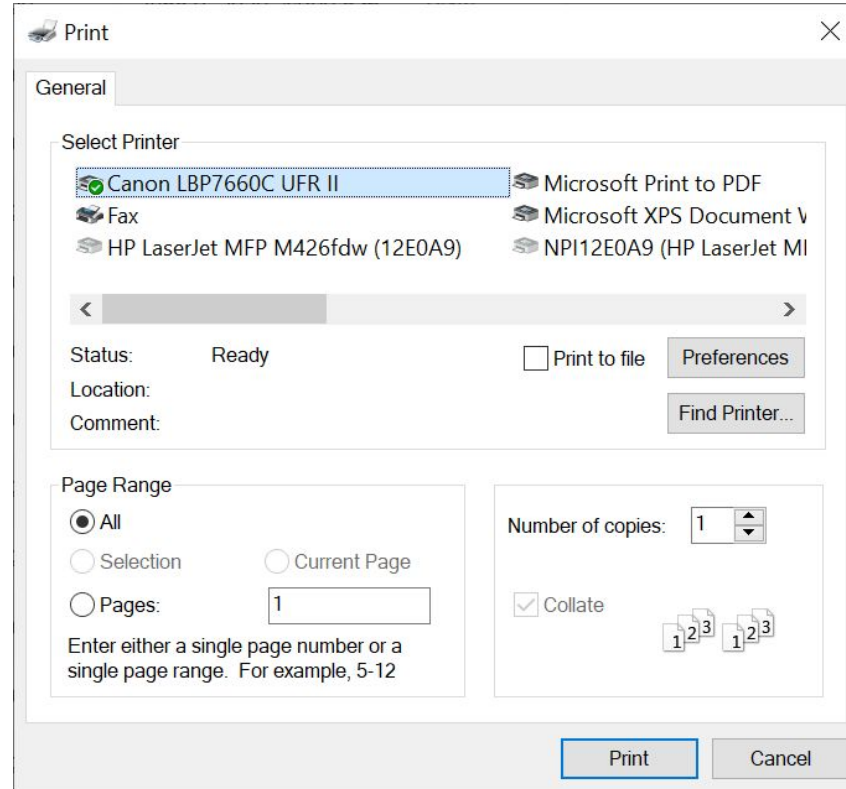
Gestalt Rule: **Emergence**



Gestalt Rule: **Closure**



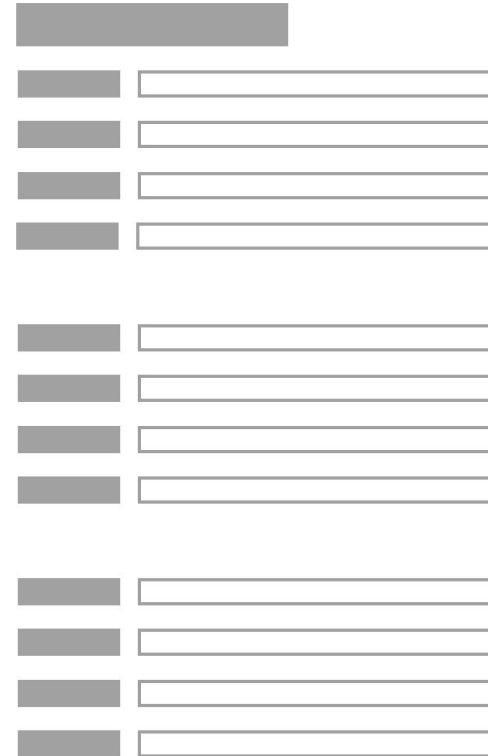
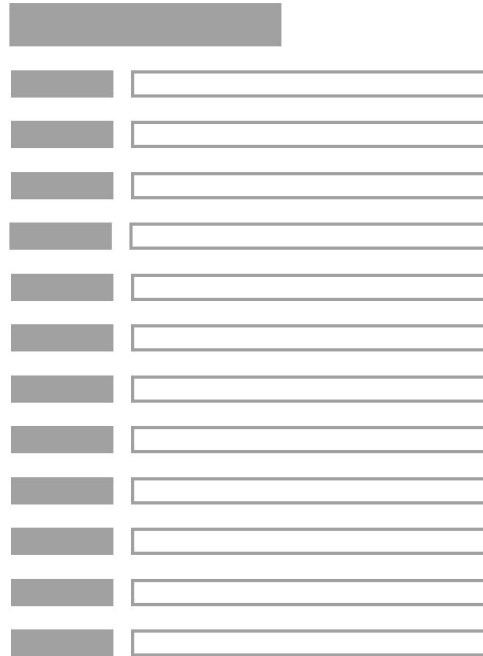
Gestalt Rule: **Common Region**



Gestalt Rule: **Continuity**



Gestalt Rule: **Proximity**



Gestalt Rule: **Multistability**





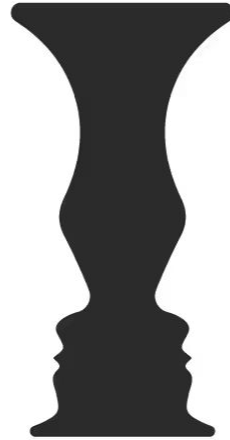


Gestalt Principles (Part 2)

- | | |
|-------------------------------|------------------------|
| 1. Emergence | 7. Figure/Ground |
| 2. Closure (Reification) | 8. Invariance |
| 3. Common Region | 9. Pragnanz |
| 4. Continuity or Continuation | 10. Similarity |
| 5. Proximity | 11. Symmetry and Order |
| 6. Multistability | 12. Common Fate |

Figure/Ground

Gestalt Rule: **Figure/Ground**



Rubin's Vase

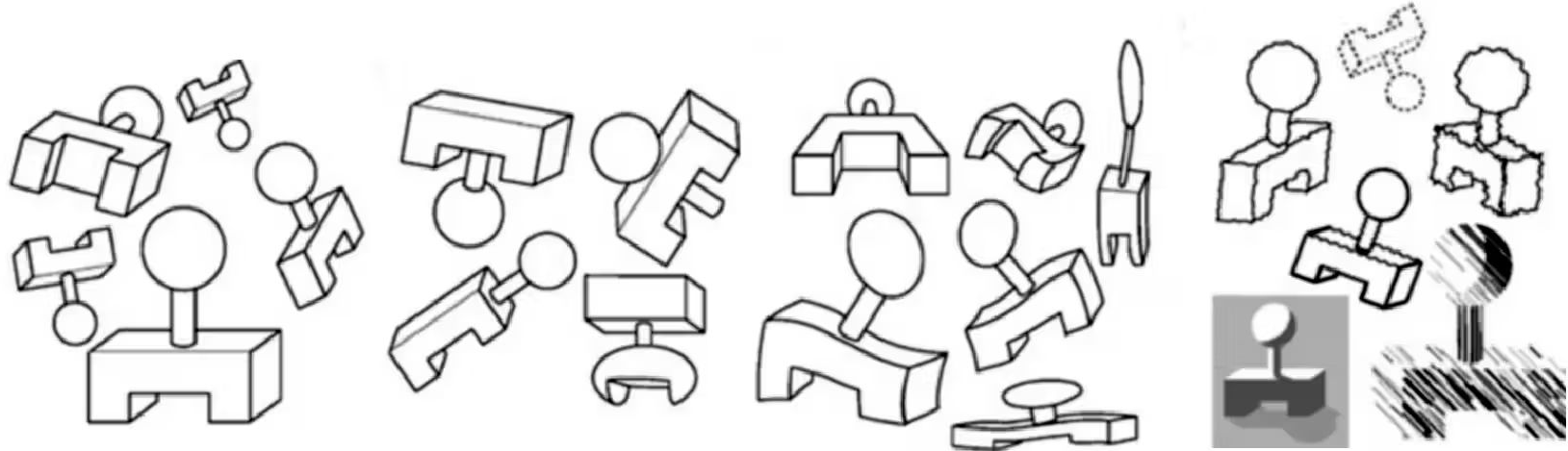


Figure/Ground

- We naturally separate visual elements into a “**figure**” (the main object) and a “**ground**” (the background). **Which element is figure can flip**, as in the vase-faces illusion.
 - Vital for understanding **how we pick out objects in a busy environment** (e.g., spotting a friend in a crowd).
 - Used in designs to create emphasis or visual interest. (e.g. infographics)
- E.g. When you read text on a page, the words are the figure and the white page is the ground. You focus on the figure (text) and let the ground (white background) fade away.

Invariance

Gestalt Rule: **Invariance**



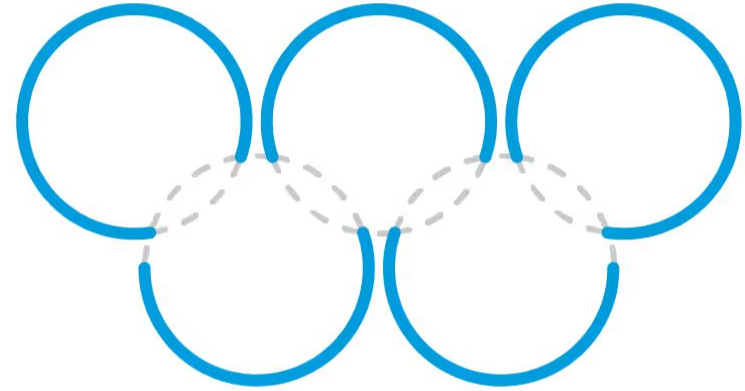
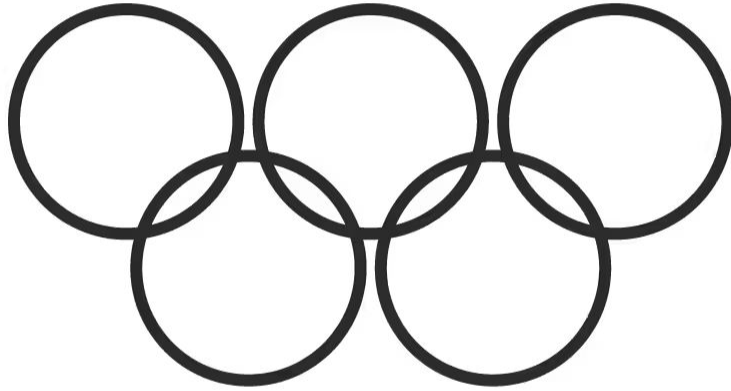


Invariance

- We can **recognize simple shapes regardless of rotation, scale, or slight distortions**. A circle is still seen as a circle even if it's tilted or partially covered.
 - Explains how we recognize letters in **different fonts** or everyday objects from various angles.
 - Helps us see objects as the “same” despite minor variations.
- E.g. Whether you see a coffee cup right-side up, upside-down, or at an angle, you still know it's a coffee cup. The shape and handle are recognizable enough.

Pragnanz

Gestalt Rule: **Pragnanz**

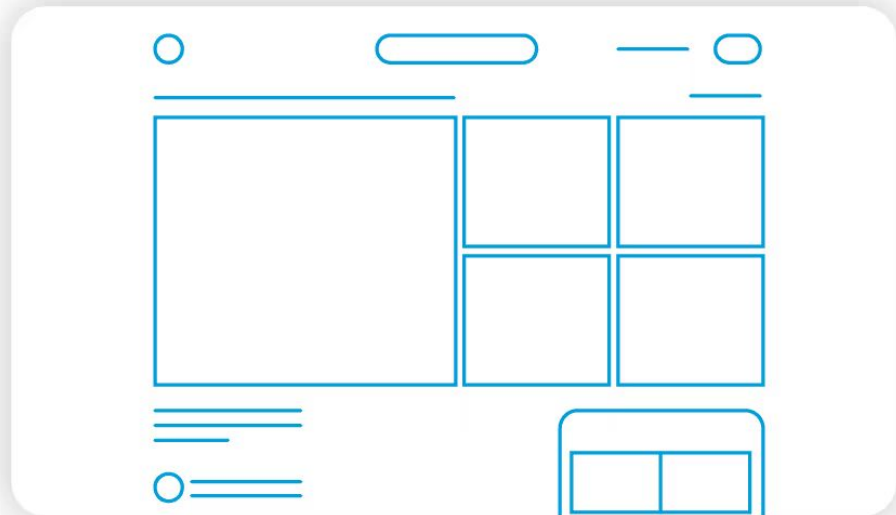
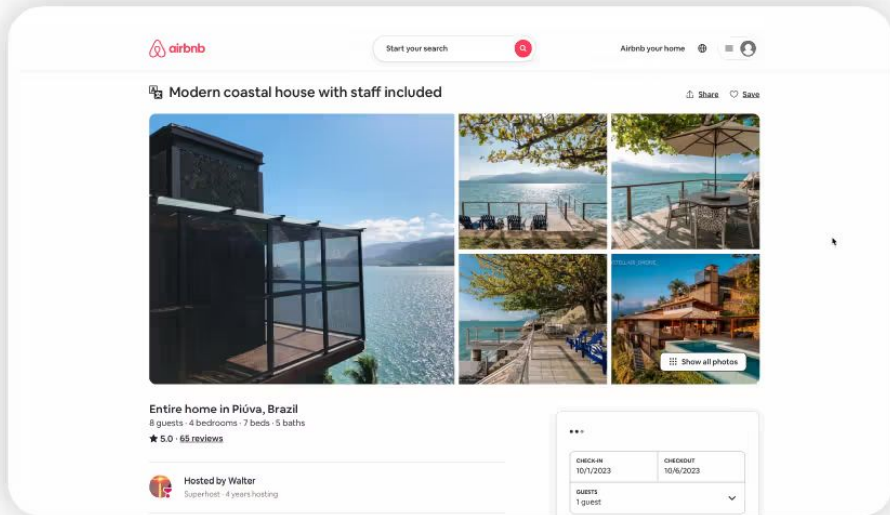




Pragnanz (German for “good figure”)

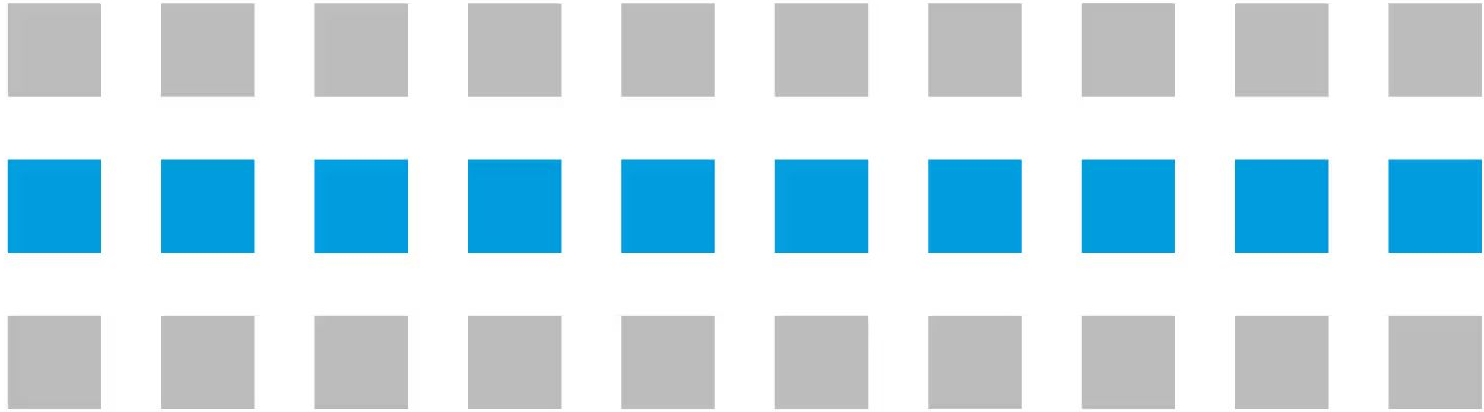
- We perceive/interpret visuals in the **simplest way possible**. Our brains automatically favor the **most stable, straightforward interpretation**.
 - Reflects our preference for symmetry, simple forms, & clear organization.
 - Guides designers to minimize visual complexity for easy understanding.
- E.g. Given a messy arrangement of coloured blocks on a table, your mind looks for basic shapes or clean lines first, rather than complex or “zigzag” patterns.

Pragnanz



Similarity

Gestalt Rule: **Similarity**





Similarity

- Elements sharing **similar characteristics** (colour, shape, size) are **mentally grouped together**.
 - Enables quick categorization based on shared features.
 - Helps you pick out familiar faces or patterns in a crowd—people wearing similar uniforms, for instance.
- A set of matching red chairs in a conference room stands out as a single group, separate from any green chairs, even if they're all arranged together in one space.



Apple Store Mac iPad iPhone Watch Vision AirPods TV & Home Entertainment Accessories Support

iPhone 16 Pro

Built for Apple Intelligence.

[Learn more](#) [Buy](#)

A side-profile view of an iPhone 16 Pro in a dark, possibly black or dark blue, finish. The phone is positioned horizontally, with its three-camera system visible on the left. Behind the phone, the word 'PRO' is rendered in large, glowing, white-outlined letters that appear to be floating or illuminated from within. The background is a dark, gradient surface.

iPhone 16

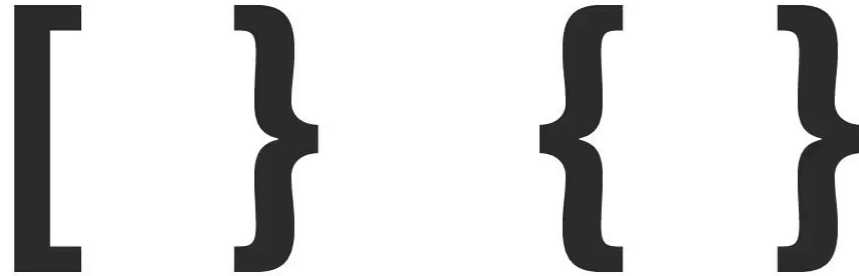
Built for Apple Intelligence.

[Learn more](#) [Buy](#)

The bottom section of the slide features a solid dark blue background. It contains the text 'iPhone 16' in a large, white, sans-serif font, followed by the tagline 'Built for Apple Intelligence.' in a smaller white font. Below the text are two white, rounded rectangular buttons: 'Learn more' and 'Buy'. At the very bottom, a thin, light blue horizontal line is visible, suggesting the top edge of the iPhone 16.

Symmetry and Order

Gestalt Rule: **Symmetry**





Symmetry and Order

- Symmetrical arrangements appear **more orderly & cohesive**. **Our brains love balance** & see symmetrical patterns as a unified whole.
 - Symmetry is often associated with beauty, harmony, and reliability.
 - Guides how we arrange items in a room or how we judge certain designs (e.g. infographics) to be aesthetically pleasing.
- A heart shape , mirrored from left to right, looks balanced and complete. You see it as a single symbol rather than two separate halves. 



Symmetry and Order

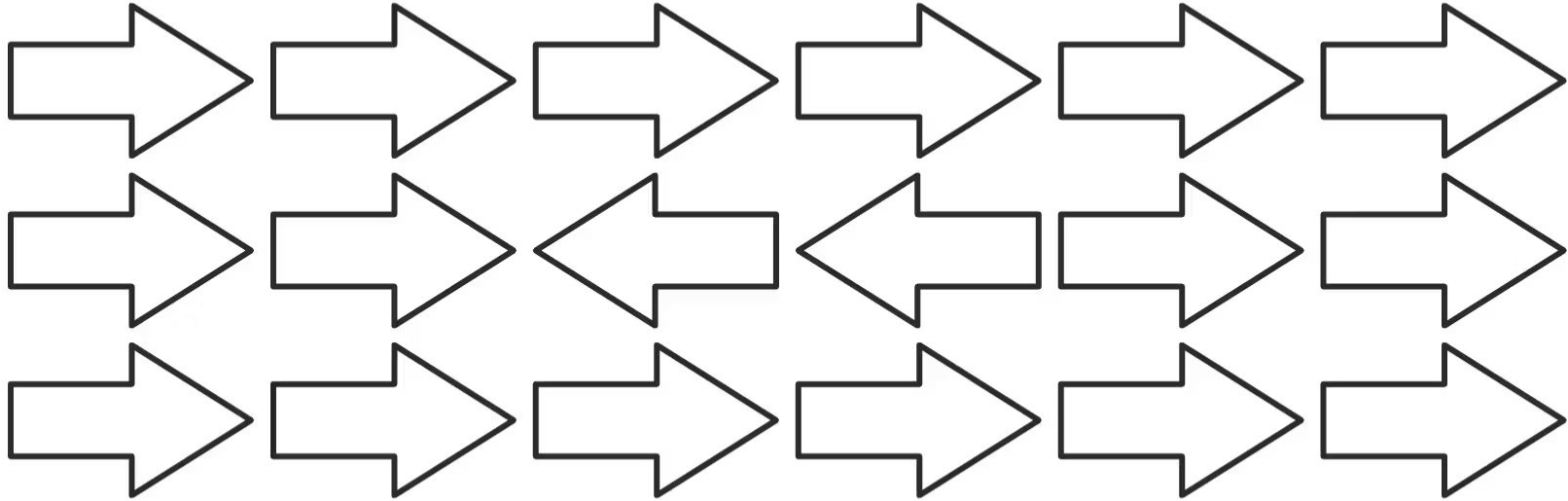
Google

Google Search

I'm Feeling Lucky

Common Fate

Gestalt Rule: **Common Fate**





Common Fate

- Elements moving in the **same direction** or showing the same change over time are perceived as **belonging together**.
 - Helps interpret motion and group dynamics in busy environments.
 - Applies to everything from following a parade to understanding traffic flows (people heading in the same direction are likely together).
- A flock of birds all heading south in formation is seen as a single group, even if they sometimes shift positions within that formation.



Common Fate

Questions related to Web Design

How to design a web page?



How much do web designers make?



How to become a web designer?



What does a web designer do?



Is web design coding?





Q: Which Gestalt Principles is most applicable to this image?



- A. Figure/Ground
- B. Invariance
- C. Pragnanz
- D. Similarity
- E. Symmetry and Order

following finding

Type the two words:

following finding

Send



Q: Which Gestalt Principles is most applicable to this image?



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Type the two words:

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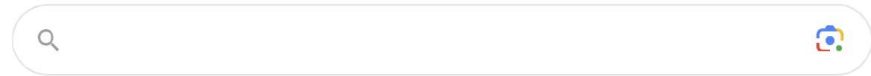
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Google Search

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I'm Feeling Lucky